An Introduction to Federated Learning and Approaches to Personalization

Vienna Deep Learning Meetup

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Institute of Science and Technology Austria (ISTA)

This Talk

1. Federated Learning and Statistical Heterogeneity.
2. Personalized Federated Learning with Hypernetworks.

What is Federated Learning?

- In standard machine learning our data is centrally accessible

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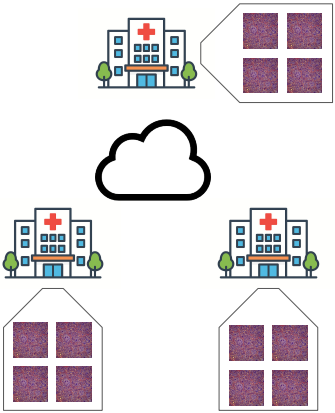
What is Federated Learning?

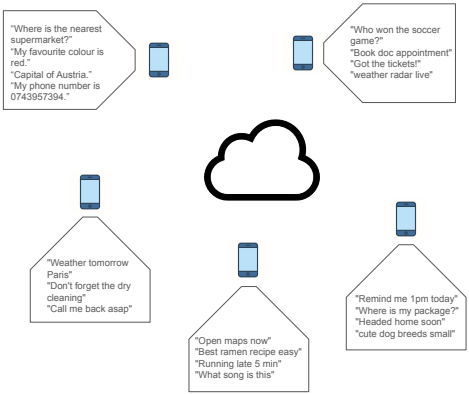
- In standard machine learning our data is centrally accessible
- Nowadays data are being generated in a decentralized manner and these are often sensitive in nature
- **Federated Learning**[1]: how can we learn from private decentralized data?

[1] McMahan et al. 2017

Federated Learning

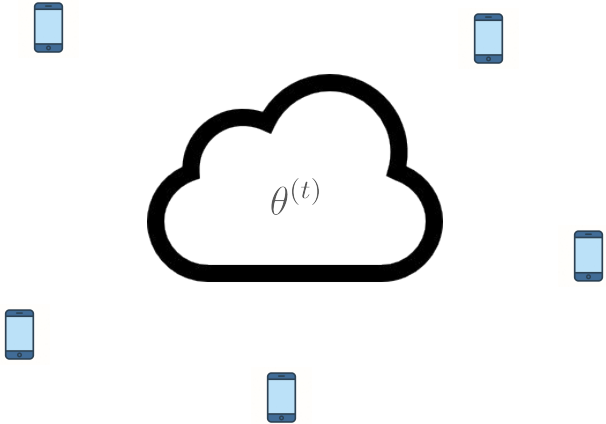


A research institution wants to train a cancer classifier on patient data in hospitals.

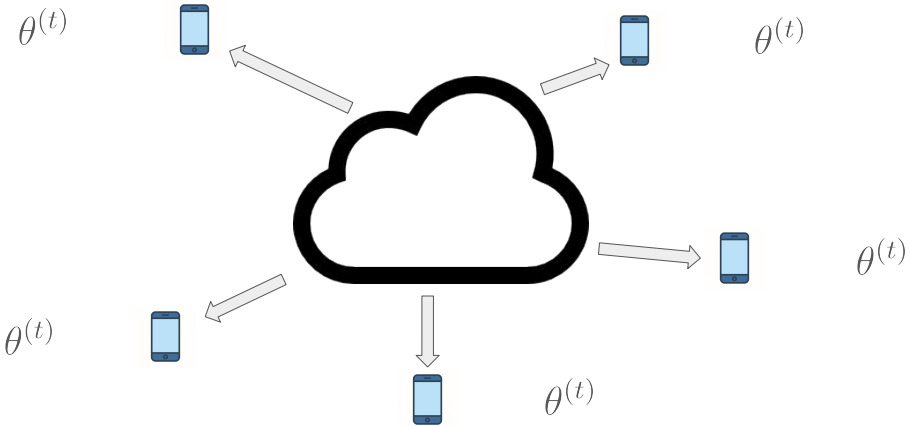


A company wants to train a next word predictor for their smartphone keyboard.

Federated Learning



Federated Learning



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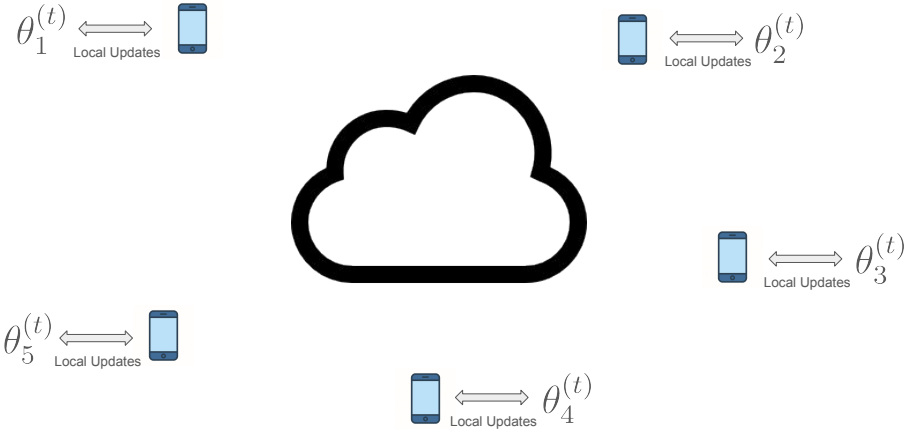

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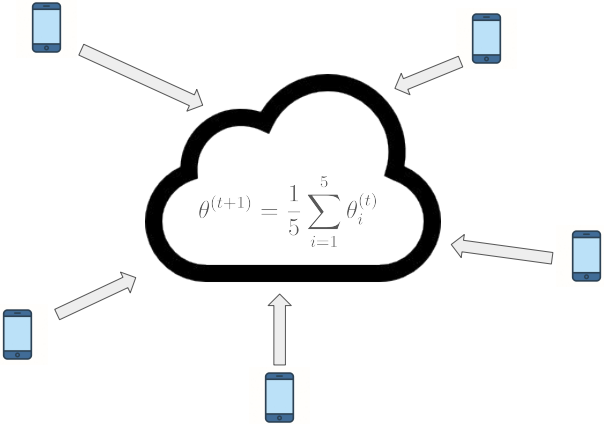

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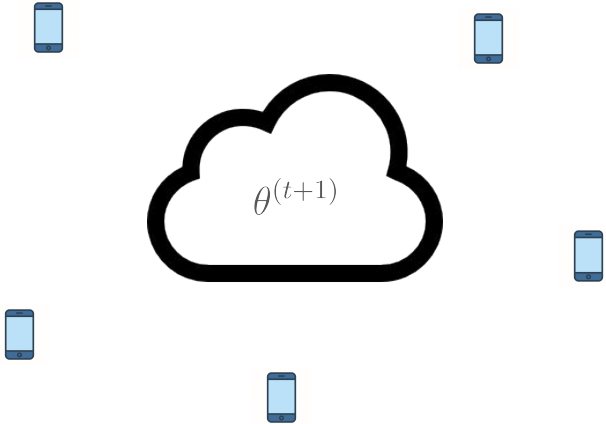

Statistical Heterogeneity

- We have a central server to coordinate training.
- We have n clients, each client i has a dataset $D_i \sim \mathcal{D}_i$ where $\mathcal{D}_i$ is a probability distribution over $\mathcal{X} \times \mathcal{Y}$.

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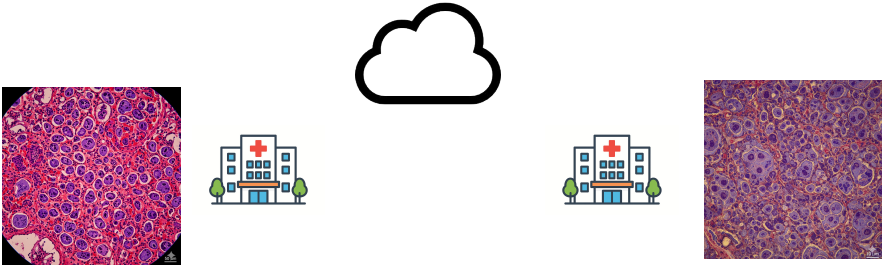
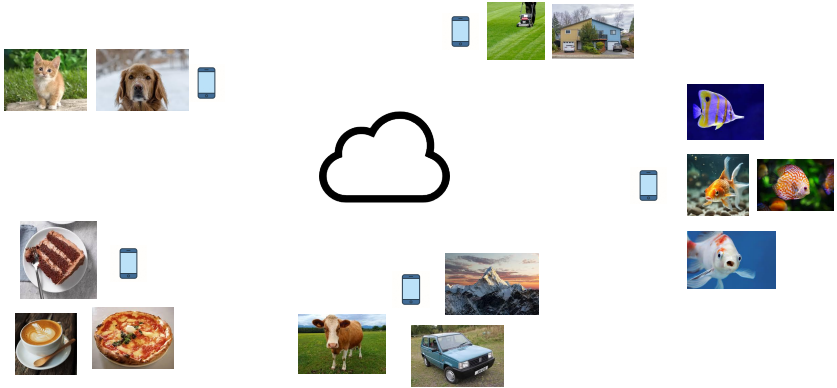
- We have n clients, each client i has a dataset $D_i \sim \mathcal{D}_i$ where $\mathcal{D}_i$ is a probability distribution over $\mathcal{X} \times \mathcal{Y}$.
- Clients are **statistically heterogeneous** if $\mathcal{D}_i \neq \mathcal{D}_j$ for $i \neq j$.

Feature Heterogeneity



The feature distributions over $\mathcal{X}$ differ between clients: $P_i(x) \neq P_j(x)$.

Label Heterogeneity



The label distributions over $\mathcal{Y}$ differ between clients: $P_i(y) \neq P_j(y)$.

Concept Shift

I just ate breakfast and I'm heading out to play …

The relationship $\mathcal{X} \to \mathcal{Y}$ differs between clients: $P_i(y \mid x) \neq P_j(y \mid x)$.

Statistical Heterogeneity

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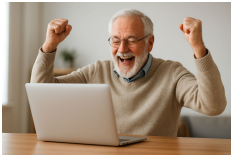
Statistical Heterogeneity

- Why is Heterogeneity a Problem?
  - Can lead to poor accuracy
  - Can make it harder to train a model
  - Can lead to biases and unfairness

Personalized Federated Learning with Hypernetworks

Training a single model might be suboptimal



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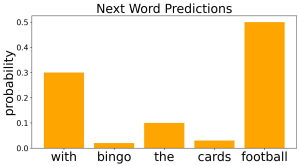

Personalized Federated Learning

- Heterogeneous client datasets $D_i \subset \mathcal{X} \times \mathcal{Y}$
- **Want to learn**: $f_i : \mathcal{X} \to \mathcal{Y}$ for each client $i \in \{1, \ldots, n\}$.
- Still make use of all client data.

Personalized Federated Learning

- Many approaches to personalized federated learning:
  - Splitting the model into shared and personalized parts[1]
  - Training local and global models[2]
  - Meta learning approaches[3]
  - Clustered federated learning[4]

[1] Collins et al. 2021
[2] T. Li et al. 2021
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[4] Ghosh et al. 2020

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- Many approaches to personalized federated learning:
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  - Clustered federated learning[4]
- **Creating a client personalized model requires training with labeled data.**

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Training free personalization

- Using training to personalize:
  - Requires labeled data
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- Using training to personalize:
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  - Can be prone to overfitting
  - More complex process, e.g. potentially adds new hyperparameters
- **How to generate a personalized model without training?**

Hypernetworks

- A neural network is a function $f(\cdot;\theta) : \mathcal{X} \to \mathbb{R}^k$, that typically directly solves the task at hand, e.g. k is the number of classes.
- A hypernetwork[1] $h(\cdot;\psi) : \mathcal{X} \to \mathbb{R}^k$ outputs the parameters to another model: k is the dimension of output model parameters.

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Idea: Use a hypernetwork to generate the personalized model parameters.

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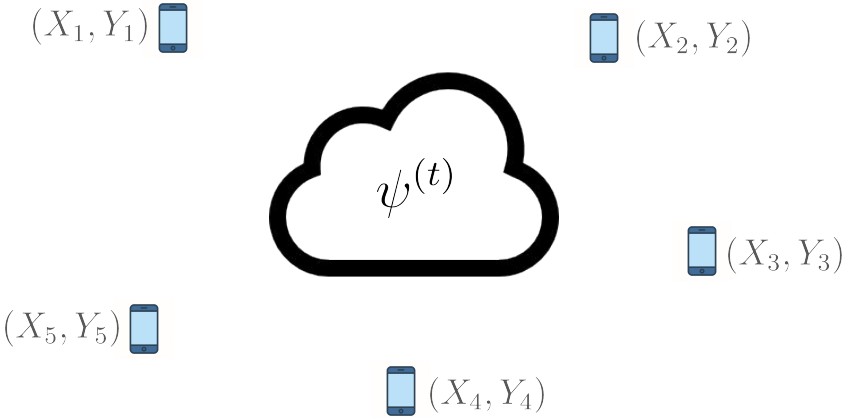
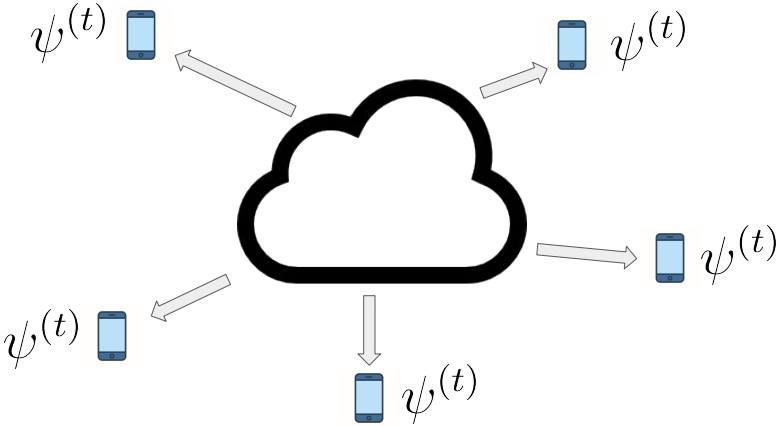
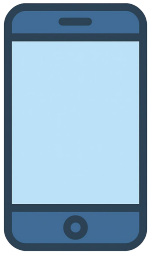
Learning the Hypernetwork Parameters



Send Parameters to Clients

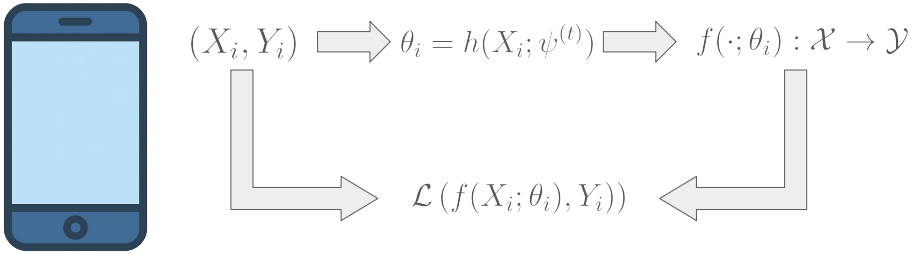


Generate Personalized Model



$$(X_i, Y_i) \Longrightarrow \theta_i = h(X_i; \psi^{(t)}) \Longrightarrow f(\cdot; \theta_i) : \mathcal{X} \to \mathcal{Y}$$

Evaluate Personalized Model



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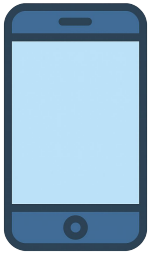
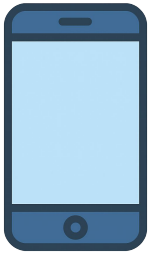


$$(X_i, Y_i) \Rightarrow \theta_i = h(X_i; \psi^{(t)}) \Rightarrow f(\cdot; \theta_i) : \mathcal{X} \to \mathcal{Y}$$

$$\mathcal{L}\left(f(X_i; \theta_i), Y_i\right)$$

Depends differentiably on $\psi^{(t)}$

Gradient Update to Hypernetwork Parameters



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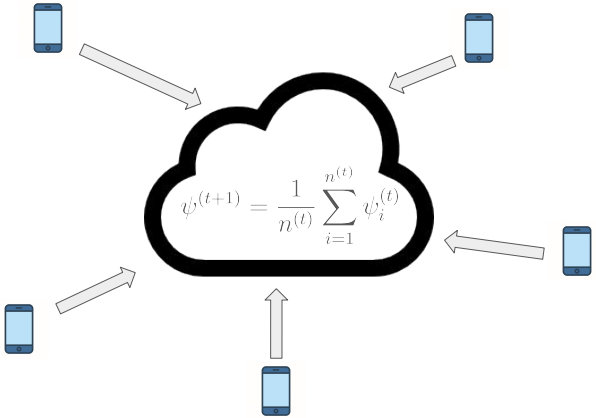
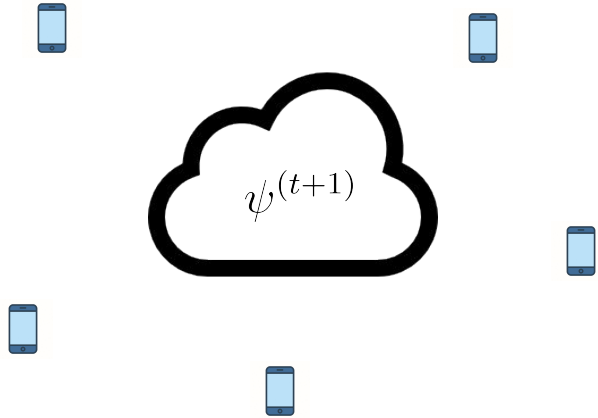
$$\mathcal{L}\left(f(X_i; \theta_i), Y_i\right)$$

$$\psi_i^{(t)} = \psi^{(t)} - \eta \nabla_\psi \mathcal{L}$$

Average Client Updates



Repeat for T Rounds



Summary so far

- We generate a personalized model with a single forward pass: $\theta_i = h(X_i; \psi)$
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- **Original Goal**: No training or labels used to personalize
- **What's the problem?**

Size of a hypernetwork

- A network $h : \mathcal{X} \to \mathbb{R}^k$ has a weight matrix in its final layer of size $n^{[L-1]} \times k$.

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Size of a hypernetwork

- A network $h : \mathcal{X} \to \mathbb{R}^k$ has a weight matrix in its final layer of size $n^{[L-1]} \times k$.
- h might be $100\times$ the size of the model we want to use
- **Too big to send to or run on the client devices.**

Hypernetworks in Federated Learning

1. Keep the hypernetwork on the server and use a more complex communication procedure[1,2]
2. Decrease the size of k using a random subspace parameterization[3]

[1] Shamsian et al. 2021
[2] Scott, Zakerinia, and Lampert 2024
[3] Zakerenia, Scott, and Lampert 2025

Random Subspace Parameterization

- For $\theta \in \mathbb{R}^d$ we can write $\theta = \theta_0 + Pv$ where
  - $\theta_0 \in \mathbb{R}^d$ is an initialization (typically random)
  - $P \in \mathbb{R}^{d \times k}$ is a random matrix
  - $v \in \mathbb{R}^k$ is now the only model parameter.

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  - $v \in \mathbb{R}^k$ is now the only model parameter.
- In a centralized setting it has been shown that k can be chosen much smaller than d while still maintaining good performance[1].

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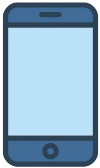
Hypernetwork + Random Subspace Parameterization



$$(X_i, Y_i) \Rightarrow v_i = h(X_i; \psi^{(t)}) \Rightarrow \theta_i = \theta_0 + P v_i \Rightarrow f(\cdot; \theta_i) : \mathcal{X} \to \mathcal{Y}$$

Experimental Setup

- We introduce different kinds of data heterogeneity:
  - Label Heterogeneity
  - Feature Heterogeneity
  - Both
- We compare to baselines that are able to create a personalized model without using labeled data

Results - Label Heterogeneity

Model	CNN			ResNet18		
p	0.1	0.2	1.0	0.1	0.2	1.0
FedAvg	47.9 ± 0.1	53.5 ± 0.4	63.6 ± 0.4	63.6 ± 0.9	68.9 ± 0.7	75.9 ± 0.3
FedProx	47.6 ± 0.3	53.3 ± 0.3	63.7 ± 0.6	63.5 ± 0.8	68.8 ± 0.5	75.9 ± 0.3
LD-FedAvg	41.6 ± 0.6	47.8 ± 1.1	56.2 ± 0.7	54.6 ± 0.8	60.0 ± 0.5	65.4 ± 0.1
FedTTA	57.2 ± 1.0	60.2 ± 1.1	69.7 ± 3.8	69.3 ± 2.3	77.2 ± 1.5	81.9 ± 2.1
ATP	53.5 ± 2.1	67.3 ± 4.3	77.8 ± 2.0	71.3 ± 2.7	**84.6 ± 1.4**	89.6 ± 0.5
FLowDUP	**66.1 ± 1.5**	**75.3 ± 1.8**	**86.6 ± 0.1**	**72.7 ± 1.3**	82.9 ± 1.8	**92.3 ± 0.4**

Table 1: Class-partitioned CIFAR10 for varying fractions, p, of the training clients having labeled data. Accuracy on unlabeled test clients.

Summary

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- A method for personalized federated learning using hypernetworks and low dimensional parameterizations

Summary

- Federated learning and statistical heterogeneity.
- A method for personalized federated learning using hypernetworks and low dimensional parameterizations
- It generates a personalized model without additional training or labels

Resources

Frameworks & Libraries:

- Apple's `pfl-research` repository: `https://github.com/apple/pfl-research`
- TensorFlow Federated: `https://www.tensorflow.org/federated`
- Flower Labs: `https://flower.ai/`

Survey:

- "Advances and Open Problems in Federated Learning", P. Kairouz et al., *arXiv:1912.04977*, 2021.

Opportunities

- For PostDoc or PhD opportunities reach out to Prof Christoph Lampert:
  - **Email:** chl@ist.ac.at
  - **Website:** `https://cvml.ista.ac.at/`
- I am looking for opportunities in industry (particularly in Vienna). Feel free to reach out or talk later!
  - **Email:** jscott@ist.ac.at
  - **Website:** `https://jonnyascott.github.io/`

Appendix

Results - Feature Heterogeneity + Both

Dataset	Rotated Fashion-MNIST			FEMNIST		
p	0.1	0.2	1.0	0.1	0.2	1.0
FedAvg	77.7 ± 0.1	79.2 ± 0.2	81.4 ± 0.6	76.5 ± 0.3	78.0 ± 0.3	84.3 ± 0.1
FedProx	77.3 ± 0.7	78.7 ± 0.4	80.0 ± 0.6	72.3 ± 0.4	74.8 ± 0.4	83.3 ± 0.0
LD-FedAvg	76.1 ± 1.1	78.8 ± 0.3	81.3 ± 0.1	61.4 ± 1.4	74.4 ± 0.3	80.6 ± 0.4
FedTTA	78.3 ± 0.5	80.2 ± 0.1	81.2 ± 0.2	71.1 ± 2.4	72.6 ± 3.5	74.3 ± 3.1
ATP	76.7 ± 0.8	79.7 ± 0.7	81.5 ± 0.7	82.9 ± 0.1	83.7 ± 0.2	84.1 ± 0.4
FLowDUP	**81.5 ± 0.3**	**83.3 ± 0.5**	**87.3 ± 0.2**	**84.5 ± 0.4**	**86.4 ± 0.3**	**89.3 ± 0.1**

Table 2: Rotated Fashion-MNIST (left) and Federated MNIST (right) for varying fractions, p, of the training clients having labeled data. Accuracy on unlabeled test clients.

Visualizing the Personalized Models

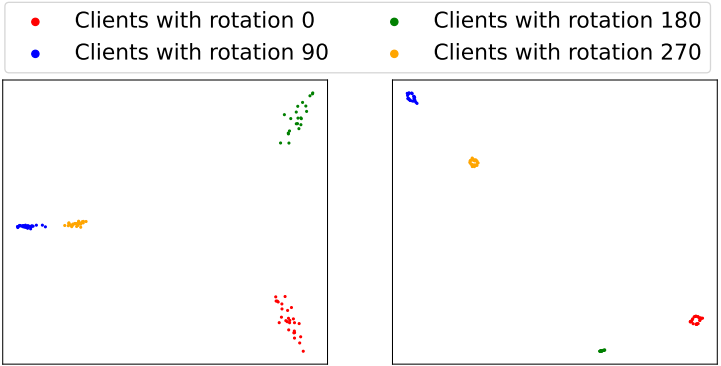

Figure 1: Visualization of the personalized models on Rotated Fashion-MNIST. Projection to two dimensions using PCA (left) and t-SNE (right).

Choice of subspace dimension k

- Varying k allows us to trade off between accuracy and communication overhead

Fraction Labeled (p)	**0.1**	**0.2**	**0.5**	**1.0**
FLowDUP ($k = 200$)	56.6 ± 1.9	65.7 ± 2.9	70.0 ± 1.8	71.2 ± 0.7
FLowDUP ($k = 500$)	60.0 ± 1.9	69.8 ± 2.8	75.9 ± 1.2	76.9 ± 0.8
FLowDUP ($k = 2000$)	64.5 ± 0.3	72.5 ± 1.7	80.3 ± 1.5	83.5 ± 0.9
FLowDUP ($k = 10000$)	66.1 ± 1.5	75.3 ± 1.8	83.3 ± 1.5	86.6 ± 0.1

Choice of network architectures

- For a fixed choice of k changing the architecture of the client model can improve performance

Model Architecture		**Fraction Labeled (p)**			
h_1	f	**0.1**	**0.2**	**0.5**	**1.0**
CNN	CNN	66.1 ± 1.5	75.3 ± 1.8	83.3 ± 1.5	86.6 ± 0.1
CNN	ResNet18	71.4 ± 0.5	80.0 ± 3.3	88.0 ± 0.9	90.7 ± 0.2
ResNet18	CNN	67.0 ± 2.4	78.0 ± 3.1	87.3 ± 1.3	88.5 ± 1.1
ResNet18	ResNet18	72.7 ± 1.3	82.9 ± 1.8	91.0 ± 0.4	92.3 ± 0.4